Project Description

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Modern AI systems rely on massive neural networks trained on static datasets. These "black boxes" fail at structural out-of-distribution generalization: when faced with a valid but never-seen combination of known concepts, they default to memorization rather than composition. They require massive data, lack interpretability, and cannot adapt without retraining.

TR-Core is a tabula-rasa symbolic cognitive architecture that autonomously discovers compositional rules from raw interaction -- without pretrained weights, embeddings, or LLMs. The system implements a Behavioral Program: a layer of changeable rules over a fixed substrate, modified through seven development operations (CREATE, MODIFY, GENERALIZE, SPECIALIZE, COMPOSE, SPLIT, DELETE).

The core innovation is an Apriori-style induction engine with an exact binomial statistical filter (alpha = 0.001, precision >= 0.95). It searches conjunctions of literals over finite attribute domains (max arity 4), using coverage-based anti-monotonicity pruning to tame combinatorial explosion. The COMPOSE operation -- conjoining rules into subprograms -- is the causal mechanism for structural-OOD generalization.

Validated on two controlled protocols: Phase 3 (8 attributes, 4,608 contexts) achieves 0.78-0.80 balanced accuracy, significantly outperforming kNN (0.70), Random Forest (0.66), and Logistic Regression (0.20). Phase 4 (Structural-OOD) tests generalization to a held-out triple "calm AND guard AND night IMPLIES approach" with zero training support; TR-Core achieves 0.96-1.00 held-out accuracy by composing the learned pair "calm AND guard IMPLIES approach". Mechanistic ablations (COMPOSE disabled, outcomes randomized) collapse to chance (0.20), isolating COMPOSE as the causal mechanism.

The system includes a Gymnasium-compatible interface, Hydra configuration, DVC pipeline, and formal complexity analysis (worst-case O(L^K), expected O((1+beta*L)^K) with exact binomial Type I error control).